Tanvir Ahmed

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Summary

I am a Robotics and Mechatronics Engineering student with practical expertise in hardware and software integration. I have hands on experience in PCB design and developing IoT enabled systems. During my university career, I have worked on various projects ranging from purely software-based applications to integrated hardware–software platforms with IoT features, as well as standalone hardware and embedded systems. I have also contributed as a research assistant, mentored students in robotics competitions, and continuously explore new technologies to expand my skill set and apply them to real-world engineering challenges. For details, click here.

Education

Bachelor of Science (B.Sc.) — Jan 2022 – present
University of Dhaka — Dhaka, Bangladesh
CGPA: 3.81 out of 4 (7th semester completed)
Major: Robotics and Mechatronics Engineering
Thesis: Design and Implementation of Cogni-Care: A Socially Assistive Robot for Patients with Mild Cognitive Impairment
Advisor: Md. Jubair Ahmed Sourov (Lecturer, RMEDU) and Dr. Sejuti Rahman (Associate Professor and Former Chairperson, RMEDU)

Higher Secondary Certificate (HSC) — 1 July, 2018 – 30 January, 2021
GPA: 5.00 out of 5.00
Dhaka Imperial College — Dhaka, Bangladesh
Stream: Science

Secondary School Certificate (SSC) — Jan 2016 – Mar 2018
GPA: 4.94 out of 5.00
Shaheen Academy School and College — Feni, Bangladesh
Stream: Science

Experience

IoT Developer — ACI PLC — May 2026 - Present

- Designing and developing a variety of connected IoT devices and embedded systems across various industries like agricultural machinery and automotive vehicles.
- I am currently responsible for building scalable hardware and cloud solutions to help in various industries and provide real-time data analytics to the users.

Undergraduate Research Assistant — Aug 2025 - April 2026

- Contributing to a funded project developing a **Socially Assistive Robot** to support patients with Mild Cognitive Impairment (MCI).
- Our goal is to develop a tabletop SAR (Socially Assistive Robot) that will interact with

the user through the display. The robot will have articulated body to make it more interactive. The robot Help MCI patients in their treatment. Record their statistics using various sensors from a smartwatch, and estimate other necessary statistics and send the data to the cloud. Based on the information stored, the robot will autonomously choose the next best exercise for the patient.

- In this project, I will be responsible for the hardware and electronics system design and implementation.
- Here, We are also Collaborating doctors to find the best possible exercises for the MCI patients. Our goal is make the whole system a cloud-based systems so that doctors can see everything one his end and give feedback. This project will reduce nurse burden and improve patients quality of life.

Undergraduate Research Assistant Dec 2023 - Aug 2025

- Contributed to developing **IHABOTv2**, a hospital robot capable of roaming hospital floors, delivering medicine to multiple patients automatically, recording vital signs, and assisting in exercise evaluation without the need for a nurse.
- Led the hardware design, integrating navigation, robotic arm, BLE sensors, and the user interface, with all systems synchronized through Google Firebase for real-time monitoring and control.
- Collaborated with a cross-functional team to seamlessly connect hardware and software, ensuring safety and efficiency of the robot.
- Helped scale the system to deliver medicine to eight patients per cycle (currently in testing), reducing nurse workload and improving delivery accuracy.

Vice President of RMEDU Student Club Sep 2024 - Sep 2025

Assisted in organizing **Robotronics Fest 2025** (Website)

- Co-led the organization of the nationwide robotics competition **Robotronics Fest 2025**, attracting over 900 participants from 50+ institutions.
- Directed the technical section for a two-day festival, managing competitions such as SoccerBot, Line Follower, Micromouse, and DU AI Challenge.
- Collaborated with faculty, sponsors, and volunteers to ensure proper execution of tasks and high engagement throughout the event.

As Vice President of the club, I also assisted in arranging other competitions and workshops:

- RMEDU Intra-Department Robotics Competition 2025: Servo Sanctum - assisted in organizing the RMEDU Intra-Department Robotics Competition 2025: Servo Sanctum alongside fellow club members.
- Seminar on Robot Operating System (ROS) - coordinated a technical seminar on ROS which was conducted by another senior.
- RMEDU Freshers' Fest 2023 - managed event activities for new students' orientation festival.
- Session on Git and GitHub - conducted a workshop on version control system for students.
- Fundraising to help flood-affected areas of the country - collaborated with teachers and fellow students to organize charity initiatives supporting flood victims.

- Workshop on Internet of Things (IoT) - Organized an hands-on IoT workshop for students of the department.

Bangladesh Robot Olympiad - BdRO Sep 2022- Apr 2024

- Collaborated with a team to organize and manage the national round of of BdRO in 2022 and 2023.
- Assisted in coordinating the participants and handling the technical and logistical tasks during competitions.
- Trained and mentored students, helping them develop robotics skills for the international competition.

Projects

IHABOTv2 Link to Video Link to Details

Led the development of Intelligent Hospital Assistant Robot Version 2

- **I Designed and developed the user interface** to monitor and control robot operations efficiently.
- Built and integrated the electronics system, ensuring reliable sensor and actuator communication.
- **Constructed the main body of the robot**, including mechanical assembly and component placement.
- I Collaborated with the autonomous navigation developer by assisting with sensor integration, testing, and debugging.

SmartBinV1 Link to Details

Personal Project, Here I collaboration with a Senior from UAP to make a smart bin that can sense the gas comming out of compostable waste.

- **Integrated the system with the cloud for real-time monitoring and data visualization through a custom-built smartphone application**, with data storage for further analysis.
- Handled sensor interfacing, data processing, and cloud communication, while collaborating on app development and hardware design.
- I also **made a crusher** for this project that crushes the trash before reaching the storage are.
- The project aimed to explore environmental monitoring and promote smarter waste management through IoT.

Real-time GPS Tracker Link to Details

Role: Hardware, Firmware, and PCB Developer

Collaborated with a senior to develop a real-time GPS tracking system using ESP and the A9G GSM/GPS module by AI Thinker.

- Designed and implemented key components of the hardware, including **designing a custom PCB** design where all other components ware seated.
- Developed firmware to handle GPS data acquisition and attempted real-time data transmission over GSM.
- I achieved success in GPS positioning (with the use of cellular network), hardware integration, even though the project was not fully completed.

Intelligent Firefighting Robot Link to Details

Large Scale Fire Fighting Robot (Personal Project)

Role: Hardware and the Software (Tkinter Based Desktop App)

- Developed a fully remote-controlled firefighting robot capable of navigating hazardous areas.
- Designed and implemented the complete software, including control algorithms and communication protocols.
- Built the entire hardware and electronics system, integrating actuators and power management for real-time operation.

AgRover – Agricultural Robot Link to Details

Course Project from Introduction to Robotics

Responsible for Hardware, Software, and Electronics Development

- Collaborated with a team to develop a prototype agricultural robot capable of autonomously navigating fields, drilling holes, and planting seeds.
- Contributed to mechanical design, electronics integration, and control logic to ensure coordinated field operations.
- Successfully completed the prototype as part of course requirements.
- Planned to scale the prototype into a real-world deployable solution to assist farmers and improve agricultural efficiency.

CMS – Class Management System Link to Details

Course Project from Object-Oriented Programming with Python

Role: User Interface Development

- Developed a cloud-based class management system using Python (OOP principles) as part of a team project.
- Implemented features to generate class routines based on teachers' availability, and manage assignments, class cancellations, and session additions.
- Designed the system to handle the end-to-end management of academic schedules, with real-time cloud synchronization to keep all users updated.

Mobile App Development

Designed and built multiple Android applications for robotics, utilities, and personal projects:

- Network Protocol Data Transfer — Simple app to communicate with a robot over the same Wi-Fi network by sending raw data to my robot. Allowing me to quickly test the device.
- To-Do Application — A personal attempt to mimic TickTick (A popular paid To-Do List and Calendar app). This early project that helped build app development fundamentals.
- TripCost — **Money management app** with expense tracking, fully **synced through Firebase Firestore** and can handle a lot of users at the same time.
- SerialTest — This is an app I build for myself to send custom serial commands via USB for microcontroller serial testing. It allows me to check whether the command is working properly or not.

Gas-Monitoring-System-IR-Based

GitHub

Here we tried to use IR module to transfer the signal from a workspace (gas environment) where there will be a lot of gas sensors. Then every sensor will transmit the IR to a main IR receiver, the main IR receive and decode the signal. The signal will be used find the specefic IR signal and see whether there are any problem or not in that are. This is still in experimental stage and we just tried to see whether this concept is actually feasible or not.

Open-Source 3D Design Contributions

I really enjoy designing 3D printable parts. For example, I created outer casings for the ESP32-CAM and ESP32-CAM-MB USB programmer, which have been downloaded over 70 times. I've also designed various other mechanical parts that help make assembling electronics easier and more fun. Sharing these designs online lets me connect with other makers and keep improving my skills.
Role: 3D Designer and 3D Printing

- GrabCAD: 3D models downloaded 5804 times (Member since June 2022).
- Cults3D: 14 designs, 593 downloads.
- Printables: Multiple models, 393 downloads, 59 likes.
- MakerWorld: 13 models, 181 downloads, 33 prints.

Awards & Activities:

- Runner-up, **Robotronics Fest 2025** (Project Showcase Section)
- 4th Place in National Round, **4th Kibo Robot Programming Challenge (Kibo-RPC)**
- Participant, **ICPC Asia Dhaka Regional** (2022, 2023, 2024) ; **BEAR Summit & National Semiconductor Symposium** (2025)
- Participant, **Idea Innovation 4.0 - Business Case Competition** by Interactive Cares (2023) and **DUEDC Skillneurs 2.0 - Business Case Competition** (2023)
- Mentored students in robotics, GitHub, and embedded system projects (2024)
- Vice President, **Robotronics Fest 2025** — organized event with 900+ participants from 50+ institutions

Technical Skills

- **Programming Languages**: Python, C, C++, Dart, C#
- **Frameworks & Libraries**: Flutter, Flet, Tkinter, CustomTkinter, NumPy, Pandas, SciPy, GeoPandas, Folium, Geemap, scikit-learn.
- **Developer Tools & Version Control**: Git, GitHub, Google Cloud, Firebase, MQTT.
- **CAD & Simulation Software**: Fusion 360, SolidWorks, FreeCAD, Ansys, EasyEDA, Tinkercad, SIMENS STAR CCM+
- **Electronics & Hardware**: Arduino, Raspberry Pi, NVIDIA Jetson, ESP32, STM32, UNIT-T tools
- **Robotics and Automation**:Niryo Robotics Arm, FESTO, Woosun, SIEMENS PLCs and SIMENS MPS

- **3D Printing Ecosystem**: Creality, FlashForge, Bambu Lab, Orca Slicer.
- **Operating Systems**: Linux (Ubuntu), Windows.

Certifications

- Using Python to Interact with the Operating System – Google/Coursera (2023)
- Crash Course on Python – Google/Coursera (2023)
- Introduction to Git and GitHub – Google/Coursera (2023)
- Troubleshooting and Debugging Techniques – Google/Coursera (2023)
- Coding Fundamentals I & II – Grasshopper (2020–2021)
- Fundamentals of Digital Marketing – Google Digital Garage (2019)
- TED-Ed Earth School for Nature – UN Environment Programme (2020)
- Workshop on IoT with Firebase and ESP32 – IEEE EDS SBC DU (2023)